

WHEELER AAF, HI, USA

WMO: 911700

Lat: 21.483N

Lon: 158.033W

Elev: 255

StdP: 98.30

Time Zone: -10.00 (W10)

Period: 06-19

WBAN: 22508

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.7	15.3	10.6	8.2	18.2	11.7	8.8	18.7	9.2	21.7	8.3	21.7	1.5	320	0.281

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.5	29.6	22.8	29.0	22.5	28.3	22.0	24.4	27.0	23.8	26.6	23.2	26.3	4.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.7	19.2	25.4	22.9	18.2	25.1	22.4	17.6	24.8	75.0	26.6	72.4	26.9	70.3	26.3	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.4	6.7	DB	12.5	30.6	0.6	1.0	12.1	31.3	11.7	31.8	11.4	32.4	11.0	33.1	(4)
(5)			WB	11.3	24.5	0.6	1.3	10.8	25.4	10.5	26.1	10.1	26.9	9.6	27.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	22.5	20.5	20.4	20.8	21.7	22.5	23.5	24.2	24.6	24.3	23.7	22.5	21.3	(6)
(7)		DBStd	1.85	1.17	1.32	1.29	1.25	0.97	0.88	1.06	1.02	1.00	0.95	1.15	1.11	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	2	1	1	1	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	4562	324	291	334	350	388	406	440	451	428	424	374	351	(10)
(11)		CDD18.3	1523	67	59	76	100	130	156	182	193	178	166	124	93	(11)
(12)		CDH23.3	7024	142	118	185	309	545	799	1088	1240	1112	847	447	191	(12)
(13)	CDH26.7	864	1	1	3	7	29	83	174	232	202	106	23	3	(13)	
(14)	Wind	WSAvg	2.8	2.5	2.7	2.9	2.9	2.8	3.0	3.0	2.7	2.5	2.6	2.5	(14)	
(15)	Precipitation	PrecAvg	1147	187	120	123	94	70	38	49	47	104	143	127	(15)	
(16)		PrecMax	1684	594	462	354	422	238	94	106	135	184	302	446	352	(16)
(17)		PrecMin	708	23	13	11	5	5	2	10	10	0	24	16	11	(17)
(18)		PrecStd	293	161	100	104	85	52	25	27	32	38	71	112	88	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.6	26.4	27.2	27.5	28.5	29.1	29.8	30.6	30.2	29.3	28.1	27.1	(19)
MCWB			20.1	20.4	20.7	20.5	21.7	22.0	22.8	23.3	23.0	23.1	22.4	20.9	(20)	
2%		DB	25.6	25.5	26.1	26.8	27.5	28.2	29.0	29.6	29.3	28.5	27.4	26.1	(21)	
		MCWB	19.8	20.0	20.2	20.3	20.7	21.4	22.5	22.8	22.7	22.7	21.8	20.6	(22)	
5%		DB	24.9	24.9	25.2	26.1	26.9	27.7	28.3	28.8	28.7	27.8	26.7	25.2	(23)	
		MCWB	19.6	19.7	19.6	20.1	20.6	20.9	22.0	22.4	22.4	22.0	21.4	20.2	(24)	
10%		DB	24.0	23.9	24.2	25.2	26.2	27.1	27.7	27.9	27.8	27.3	26.0	24.4	(25)	
		MCWB	19.3	19.3	19.3	19.9	20.3	20.7	21.5	21.6	21.5	21.7	21.0	20.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.8	22.5	22.3	22.7	22.9	24.1	24.6	25.2	24.9	25.0	24.1	22.5	(27)
MCDWB			24.4	23.9	25.1	24.6	26.4	25.7	27.2	28.1	27.6	27.1	26.2	24.5	(28)	
2%		WB	21.1	21.4	21.3	21.7	22.1	22.8	23.6	24.1	24.1	23.9	23.0	21.9	(29)	
		MCDWB	23.7	23.7	24.3	24.4	25.8	26.0	26.6	27.2	26.8	26.6	25.5	24.3	(30)	
5%		WB	20.6	20.8	20.6	21.1	21.6	22.0	22.9	23.5	23.4	23.2	22.5	21.4	(31)	
		MCDWB	23.2	23.2	23.6	24.1	25.3	26.0	26.4	26.8	26.6	26.2	25.1	23.7	(32)	
10%		WB	20.0	20.2	20.0	20.6	21.1	21.5	22.4	22.8	22.8	22.7	21.8	20.9	(33)	
		MCDWB	22.9	22.8	23.1	23.8	24.6	25.6	26.1	26.3	26.3	25.8	24.6	23.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.6	7.2	7.2	7.1	7.6	7.4	7.3	7.5	7.8	7.5	7.0	6.5	(35)
MCDBR			8.7	8.4	8.5	8.2	8.5	8.1	7.8	8.2	8.6	8.3	8.0	7.7	(36)	
MCWBR			4.4	4.3	4.0	3.7	3.7	3.1	3.0	3.1	3.4	3.6	3.8	3.8	(37)	
5% WB		MCDBR	7.7	7.2	7.8	7.2	7.8	7.6	7.0	7.4	7.1	7.6	7.3	6.6	(38)	
		MCWBR	4.3	4.1	4.1	3.6	3.6	3.2	3.0	3.3	3.3	3.7	3.7	3.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.344	0.350	0.357	0.374	0.368	0.357	0.356	0.355	0.348	0.358	0.356	0.340	(40)	
taud		2.615	2.586	2.573	2.506	2.556	2.596	2.611	2.614	2.638	2.617	2.613	2.648	(41)		
Ebn at Noon		932	945	947	928	924	926	928	934	943	925	912	925	(42)		
Edh at Noon		88	96	101	110	104	99	97	97	93	92	88	83	(43)		
(44)	All-Sky Solar Radiation	RadAvg	4.42	4.97	5.71	6.45	6.91	7.23	7.11	6.82	6.38	5.36	4.52	4.08	(44)	
(45)		RadStd	0.19	0.35	0.60	0.36	0.40	0.20	0.15	0.27	0.19	0.17	0.12	0.29	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+68	+68

Nomenclature: See separate page